Alphabetical Glossary

Every important term, one line each: definition, then practical meaning.

Adjusted R^2 — R^2 penalized for the number of predictors used. *Practical meaning:* prevents rewarding a model just for adding more features.

Bias — Error from overly simplistic assumptions in the model. *Practical meaning:* high bias shows up as underfitting.

Coefficient — A number multiplying a feature in the regression equation. *Practical meaning:* its size and sign describe that feature's estimated effect.

Data leakage — Test-set information improperly influencing training. *Practical meaning:* always fit preprocessing on the training set only.

Feature — An input variable used to make a prediction. *Practical meaning:* what goes in on the x-side of the equation.

Heteroscedasticity — Non-constant variance of residuals. *Practical meaning:* shows up as a funnel shape in residual plots.

Homoscedasticity — Constant variance of residuals. *Practical meaning:* the healthy, assumption-satisfying opposite of heteroscedasticity.

Influential point — An observation whose removal meaningfully changes the fitted line. *Practical meaning:* worth investigating individually, not just averaging away.

Intercept — The predicted value when all features equal zero (b_0). *Practical meaning:* positions the line vertically; often not literally meaningful.

Lasso — Regression with an L1 (absolute-value) coefficient penalty. *Practical meaning:* can zero out irrelevant features automatically.

Leverage — How unusual an observation's feature values are. *Practical meaning:* high-leverage points have more potential to influence the fit.

MSE — Mean Squared Error, the average squared residual. *Practical meaning:* the quantity OLS directly minimizes.

Multicollinearity — High correlation among predictor features. *Practical meaning:* destabilizes individual coefficient estimates.

OLS — Ordinary Least Squares, the method of finding coefficients that minimize SSE. *Practical meaning:* the default fitting method for standard linear regression.

Outlier — A data point far from the general trend. *Practical meaning:* can disproportionately affect a squared-error model.

Overfitting — Fitting training data (including its noise) too closely. *Practical meaning:* good training performance, poor generalization.

Prediction ($\hat{y}$) — The model's output for a given input. *Practical meaning:* what you actually deliver to a downstream system or user.

R^2 — Proportion of variance explained relative to the mean baseline. *Practical meaning:* a normalized, unitless goodness-of-fit summary.

Regularization — Adding a penalty for coefficient size to the loss function. *Practical meaning:* controls model complexity and helps generalization.

Ridge — Regression with an L2 (squared) coefficient penalty. *Practical meaning:* shrinks coefficients smoothly without zeroing them out.

RMSE — Square root of MSE. *Practical meaning:* error in the same units as the target, still sensitive to large errors.

Slope — How much the prediction changes per one-unit increase in a feature (b_1). *Practical meaning:* the direct, interpretable "effect size" of a feature.

SSE — Sum of Squared Errors. *Practical meaning:* the raw quantity OLS minimizes before averaging.

Target — The variable being predicted. *Practical meaning:* the y-side of the equation.

Test set — Data held back purely for final evaluation. *Practical meaning:* your honest estimate of real-world performance.

Training set — Data used to fit the model's coefficients. *Practical meaning:* everything the model is allowed to "see" and learn from.

Underfitting — A model too simple to capture the real pattern. *Practical meaning:* poor performance on both training and test data.

Validation set — A held-out split used for tuning decisions before final testing. *Practical meaning:* keeps the test set untouched until the very end.

Variance — Error from sensitivity to small fluctuations in training data. *Practical meaning:* high variance shows up as overfitting.